

# The Literature Review

120 minutes · 5 self-check questions

Author: **Jesse Anak** · Institute of Computational Biology and Bioinformatics (ICBB)

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*What a literature review is for, the main types and when each is appropriate, and how to build one that argues rather than merely lists.*

## Learning objectives

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- Explain the four functions a literature review performs in a research project
- Distinguish narrative, scoping, systematic, rapid and integrative reviews
- Select the review type appropriate to your purpose and resources
- Organise a review thematically rather than source by source
- Use a synthesis matrix to move from reading to argument
- Recognise and avoid the most common review failures

## What the review is for

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- To establish what is already known, so you do not repeat settled work
- To locate and justify the gap your study will fill
- To inform your method — which designs, instruments and measures others found workable
- To supply the framework against which you will later interpret your own findings
- A review is an argument leading to your question, not a catalogue of everything published

### IN PRACTICE

The single most common examiner criticism is a review that summarises twenty studies in twenty paragraphs and never says what they collectively mean. If a paragraph begins with an author name rather than an idea, suspect this failure.

## Types of review

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- Narrative review — broad, expert overview; flexible but vulnerable to selection bias
- Scoping review — maps the extent and nature of evidence and identifies gaps; ideal when a field is new or heterogeneous
- Systematic review — answers a focused question using a pre-registered protocol and exhaustive search; the highest standard, and the most work

- Meta-analysis — statistical pooling of results across studies; only valid when studies are sufficiently comparable
- Rapid review — systematic methods with deliberate, documented shortcuts, used under time pressure
- Integrative review — combines experimental and non-experimental evidence, common in nursing and health services research

#### *Choosing a review type*

| If your purpose is...                   | Use                                  | Typical effort |
|-----------------------------------------|--------------------------------------|----------------|
| Introduce a thesis chapter              | Narrative or integrative             | 4-8 weeks      |
| Map an emerging or scattered field      | Scoping review                       | 3-6 months     |
| Answer a focused effectiveness question | Systematic review                    | 6-12 months    |
| Pool effect estimates numerically       | Systematic review with meta-analysis | 9-18 months    |
| Inform an urgent policy decision        | Rapid review                         | 4-12 weeks     |

## Building the review

- Define the scope first: concepts, populations, date range, languages and study types, with a reason for each limit
- Search systematically even for a narrative review — record databases, terms and dates so the search can be repeated
- Screen in two passes: title and abstract, then full text against explicit criteria
- Extract into a synthesis matrix: rows are studies, columns are the themes, methods and findings you care about
- Reading the matrix down the columns rather than across the rows is what turns reading into synthesis
- Organise the written review by theme, debate or chronology of ideas — never by author

## Appraising what you find

- Not all evidence carries equal weight; state how you judged quality
- Evidence hierarchies place systematic reviews and randomised trials above cohort, case-control and cross-sectional designs for effectiveness questions
- Hierarchies apply to questions of effectiveness, not to questions of meaning or experience — a qualitative study is not "low quality" evidence for how patients understand illness
- Use a structured tool: CASP checklists, the Newcastle-Ottawa Scale, or Cochrane Risk of Bias 2

- Watch for predatory journals — check indexing, editorial board, and whether the publisher appears in DOAJ or on Cabells lists
- Note conflicts of interest and funding sources

## Common failures

- Listing rather than synthesising — the "annotated bibliography" review
- Relying only on what is freely available, producing a review biased towards open-access journals
- Citing abstracts without reading the full paper, and repeating a misreading from a secondary source
- Ignoring evidence that contradicts your position, which reviewers will notice
- Reviewing only recent work and losing the origin of a debate
- Letting the review stop when data collection begins — it must be updated before submission

## Worked examples

### Turning five papers into one paragraph

*A student has read five studies on adherence to hypertension medication and has written five paragraphs, one per study, each beginning with an author name.*

1. Build a matrix: rows are the five studies, columns are design, setting, sample, adherence measure and main finding.
2. Reading down the "adherence measure" column shows three used self-report and two used pill counts.
3. Reading down "main finding" shows the self-report studies all report higher adherence than the pill-count studies.
4. That comparison is the paragraph: "Reported adherence varies with how it was measured, with self-report consistently higher than objective measures (refs) — a pattern that complicates comparison across settings."
5. No sentence in that paragraph has an author as its subject. The studies are evidence for a claim, not the topic of the writing.

**The point:** Synthesis is what you see reading down the columns. Summary is what you get reading across the rows. Reviewers can tell instantly which one you did.

## Choosing the review your timeline can survive

*A student with nine months until submission proposes "a systematic review and meta-analysis of interventions to reduce malaria in under-fives across sub-Saharan Africa".*

1. A proper systematic review needs a protocol, two independent screeners, dual data extraction and risk-of-bias assessment.
2. That question would return several thousand records to screen at title and abstract.
3. She is working alone, so the second screener does not exist — and without one it is not a systematic review by PRISMA standards.
4. A scoping review of the same area is defensible, achievable solo, and still publishable.
5. Alternatively, narrow the question hard: one intervention, one age band, one country — which can make a systematic review feasible.

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**The point:** Pick the review type from your resources and question, not from which label sounds most impressive. A finished scoping review beats an abandoned systematic one.

## Key terms

### Synthesis matrix

A table with studies as rows and themes or attributes as columns, used to compare across sources rather than summarise each in turn.

### Scoping review

A structured review that maps the extent, range and nature of evidence on a topic and identifies gaps.

### Grey literature

Material outside commercial publishing — theses, government and NGO reports, working papers, conference abstracts.

### Publication bias

The tendency for studies with positive or significant results to be published more often, distorting the apparent evidence.

### Predatory journal

A journal charging publication fees while providing little or no genuine peer review or editorial service.

### Saturation (in reviewing)

The point at which additional searching returns no new relevant concepts.

## Activity

Take five papers you have already read. Build a synthesis matrix with columns for design, sample, setting, key finding and limitation. Then write one paragraph that compares across the column "key finding" without naming an author as the subject of any sentence.

## Further reading

- **Booth, Sutton & Papaioannou — Systematic Approaches to a Successful Literature Review** — The best single book on choosing and executing a review type.
- **Arksey & O'Malley (2005) — scoping review framework** — The foundational scoping methodology, refined later by Levac and by JBI.
- **Page et al. (2021) — PRISMA 2020 statement** — The reporting standard; read it before you start searching, not after.

## Self-check questions

1. A review that maps how much and what kind of research exists on a broad, emerging topic, without pooling results, is a:
  - a. Meta-analysis
  - b. Scoping review
  - c. Rapid review
  - d. Narrative review
2. The clearest sign that a literature review is listing rather than synthesising is that:
  - a. It cites more than fifty sources
  - b. Most paragraphs begin with an author name and describe one study each
  - c. It includes grey literature
  - d. It is organised chronologically
3. Judging a qualitative study of patient experience as "weak evidence" because it sits low on the evidence hierarchy is:
  - a. Correct — hierarchies apply to all research
  - b. A misapplication, because hierarchies rank evidence for effectiveness questions
  - c. Correct only if the sample is under 30
  - d. Correct because qualitative work is not generalisable
4. Excluding studies that are not freely available online risks introducing:

- a. Recall bias
  - b. Access bias, skewing the review towards open-access sources
  - c. Confounding
  - d. Attrition bias
5. When should the literature review be considered finished?
- a. When the proposal is approved
  - b. When data collection begins
  - c. Shortly before submission, after a final updating search
  - d. After fifty sources have been cited
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## Answers

1. **b.** Scoping reviews are designed to map extent, range and nature of evidence and to identify gaps. They deliberately do not pool effects or usually appraise quality in the way a systematic review does.
  2. **b.** Paragraph-per-study structure signals summary rather than synthesis. A synthesising review organises paragraphs around ideas and cites several studies within each to support, qualify or contest a claim.
  3. **b.** Evidence hierarchies rank designs for answering questions about whether an intervention works. They say nothing about the quality of evidence for questions of meaning, experience or process, where qualitative designs are the appropriate choice.
  4. **b.** Restricting to what you can download for free is a selection problem. Use your institutional library, interlibrary loan, HINARI or AJOL, or write to the corresponding author, and state any unavoidable restriction explicitly.
  5. **c.** The field continues publishing while you work. A final updating search before submission protects you from the examiner who knows of a directly relevant paper published during your fieldwork.
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